CO1 - AT2

Constraint - Based Problem Solving

Name : Vinil Kumar. S
Regno : 192511226
Sub : Artificial intelligence
Subcode : CSA1710

1. Hospital doctor - Shift Assignment Using Backtracking Search

Three doctors (D1, D2, D3) must be assigned to morning, Afternoon, and Night Shifts. Each doctor gets exactly one Shift, and each shift has only one doctor.

Constraints :

- D1 Cannot work night
- D2 must work before D3.
- D3 Cannot work morning
- No two doctors can share a shift.

Backtracking Search and Intermediate Steps

Backtracking assigns shifts one by one and rejects assignments that violate constraints.

Step	Assignment	Result
1.	D1 = Morning	Accept
2.	D2 = Morning	Reject (shift occupied)
3.	D2 = Afternoon	Accept
4.	D3 = Morning	Reject (not allowed)
5.	D3 = Afternoon	Reject (shift occupied)
6.	D3 = Night	Accept.

Final Valid Schedule

Shift	Doctor
Morning	D1
Afternoon	D2
Night	D3

Backtracking is suitable because it assigns values step by step, check constraints, and rejects invalid choices. The final schedule satisfies all given constraints.

---

2. Robot Navigation using BFS

A robot must travel through a 5×5 grid from start (S) to Goal (G). It can move up, Down, left and right.

For demonstration, assume S = (1,1), G = (5,5) and obstacles at (2,2), (2,3), (4,3), (4,4) since the question does not specify their positions

2. Manhattan distance

The heuristic estimates the distance to the goal:

$$h(n) = |r_n - r_G| + |c_n - c_G|$$

At the starting position:

$$h(S) = |1-5| + |1-5| = 8.$$

BFS Search Steps :-

BFS uses a FIFO queue and explores neighbouring cells level by level.

Step	Node explained	New nodes added
1	(1,1)	(1,2), (2,1)
2	(1,2)	(1,3)
3	(2,1)	(3,1)
4	(1,3)	(1,4)
5	(1,4)	(1,5)
6	(1,5)	(2,5)
7	(2,5)	(3,5)
8	(3,5)	(4,5)
9	(4,5)	(5,5), Goal.

Blocked cells and already visited cells are not added to the queue.

Optimal path & cost :-

One shortest path is : $(1,1) \rightarrow (1,2) \rightarrow (1,3) \rightarrow (1,4) \rightarrow (1,5) \rightarrow (2,5) \rightarrow (3,5) \rightarrow (4,5) \Rightarrow (5,5)$.

Total cost : $\text{Cost} = 8 \times 1 = 8$

BFS finds the shortest path when all moves have equal cost. Manhattan distance estimates the remaining distance ; but standard BFS does not use it to prioritize nodes ; A* is the algorithm that uses this heuristic for informed search.

Constraint analysis :

The robot must reach a survivor while avoiding obstacles & minimizing cost.

Constraint	Effect
Four-direction movement	Limits available actions.
Obstacle	Prevent movement through blocked cells
Limited Sensing	Only adjacent cells can be observed
Movement cost = 1	Each normal move adds one unit
Risk penalty = 2	Entering a risky cell costs 3 in total.
Dynamic knowledge	The robot must update its map & replan.

Solutions using Uniform cost search :-

Uniform cost search (UCS) expands the cell with the lowest accumulated path cost. An online search agent helps the robot explore unknown areas and update its map.

Movement cost :

$$C(s, s') = \begin{cases} 1, \text{ safe cell} \\ 3, \text{ risky cell} \\ \infty, \text{ blocked cell.} \end{cases}$$

Steps :

1. Start at S and Observe adjacent cells.
2. Update the map with known obstacles and risky areas

3. Add reachable cells to a priority queue according to Path cost.

4. Expand the lowest-cost cell.

5. Replan if new obstacles or hazards are discovered.

6. Continue until Gr is reached.

Application of AI concepts :-

1. Intelligent agent : The robot Senses nearby cells, makes decisions, and moves toward the survivor.

2. Rationality : It chooses actions that minize cost while avoiding known obstacles and reducing risk, based on available information.

3. Environment Type :

- partially Observable : only adjacent cells are visible
- Discrete : The building is represented as grid cells.
- Dynamic : Hazards or available paths may change.
- Single-agent : One robot makes decisions.
- Initially Unknown : The robot learns the environment through Sensing.

Uniform Cost Search (UCS) is suitable because movement costs differ between safe and risky cells. It finds a minimum cost route on the current known map.

Thus, Combining UCS with Online Search Supports safe, adaptable, and cost-effective rescue navigation.